Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

Design and Analysis of Algorithms

CSIT — 5th Semester — 2080 — Answer Sheet

Subject Code: CSC325

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What is recurrence relation? How it can be solved? Show that time complexity of the recurrence relation $T(n) = 2T(n/2) + 1$ is $O(n)$ using substitution method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Write down advantages of dynamic programming over greedy strategy. Find optimal bracketing to multiply 4 matrices of order {2, 3, 4, 2, 5}.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Discuss heapify operation with example. Write down its algorithm and analyze its time and space complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Define RAM model. Write down iterative algorithm for finding factorial and provide its detailed analysis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Write down algorithm of insertion sort and analyze its time and space complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write down minmax algorithm and analyze its complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. When greedy strategy provides optimal solution? Write down job sequencing with deadlines algorithm and analyze its complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Suppose that a message contains alphabet frequencies as given below, and find Huffman codes for each alphabet.SymbolFrequencya30b20c25d15e35

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Does backtracking give multiple solution? Trace the subset sum algorithm for the set {3, 5, 2, 4, 1} and sum = 8.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Why extended Euclidean algorithm is used? Write down its algorithm and analyze its complexity.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define NP-complete problems with examples. Give brief proof of the statement "SAT is NP-complete".

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short Notes ona) Aggregate Analysisb) Selection problems

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
